MULTI-SCALE GEM POOLING WITH N-PAIR CENTER LOSS FOR FINE-GRAINED IMAGE SEARCH

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ABSTRACT

Most existing fine-grained image retrieval schemes are built based upon deep feature learning paradigms, which typically leverage the feature maps of the last convolutional layer as features. However, such representation focuses only on the global information of the object, leaving the local details unexploited, which is however crucial to identifying subtle differences for fine-grained retrieval. In this paper, we have discovered that the mid-level feature map roles as local salient regions, which well complements the existing global feature representations. To this end, a multi-layer framework is proposed to integrate both local and global representations with generalized mean (GeM) pooling and attention mechanism, trained with the proposed N-pair Center loss to learn more discriminative features. By doing so, state-of-the-art performance can be achieved without using the hard or negative example minings. In the experiments, our approach outperforms favourably compared to the current state-of-the-art methods on the CUB-200-2011, CARS196 and In-shop Clothes Retrieval datasets.

***Index Terms*—** Deep feature learning, Attention mechanism, Fine-grained image search

1. INTRODUCTION

Image retrieval [1, 2, 3, 4, 5] has been a long standing research topic in the computer vision society. Under this topic, Fine-Grained Image Search (FGIS) was first proposed in [6] and then became a research hotspot [7, 8, 9, 10, 11] recently. Given a query image, FGIS aims to retrieve the images with same category (*e.g. Groove billed ani*) rather than other similar categories (*e.g. Brewer blackbird*) from the standard fine-grained retrieval dataset (*e.g. CUB-200-2011 [12]*). The challenge of this problem is to design robust representations which can discriminate the difference among almost identical categories. Recently, with the resurgence of deep learning [13, 14], the methods represented by deep feature learning (DFL) are the mainstream of the cutting-edge image retrieval. In a standard DFL framework, given the input label

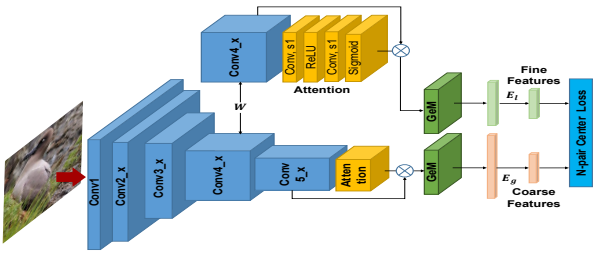

Fig. 1. The proposed framework. These features extracted from conv5_x and conv4_x are trained with the N-pair Center loss and each block followed by an attention module. The final representation is the concatenate of the coarse features and the fine features, based on which we retrieve the top-K related images employing the Euclidean distance.

annotations, deep convolutional neural networks are directly utilized to learn a set of nonlinear transformations f projecting the input image x into a lower dimensional feature space $f(x)$, which makes intra-class features have smaller distances than inter-class features. Mainstream methods focus on the DFL based on global features and effective loss functions, such as Triplet loss [15], N-pair loss [16], CRL loss [17], HDC [18] etc. Triplet loss minimizes the distance between an anchor and a positive in the feature space, both of which have the same identity, and maximizes the distance between the anchor and a negative of different identity. Since massive triplets are not helpful in training, the triplet loss needs to take a lot of time to choose hard triplets in the training process. To alleviate this problem, the N-pair [16] loss allows to jointly compare more negative samples so as to leverage more information in a batch. In order to accelerate training convergence, CRL loss [17] directly utilizes the mean value of each class in a batch to construct triplets. However, these methods do not consider about how to embed local representation into feature learning. A few methods take account of the fusion of local and global features. HDC [18] aims to mine hard examples at multi-scale by integrating a set of models with different complexity in cascade mode, however, it is inefficient to select negative samples.

In deep residual networks, we notice that the feature maps of the last block concern on the global information of the object while the outputs of penultimate block pay more at-

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tention to some parts of the target. As shown in Fig. 2, it is obvious that the heatmap of conv5_x covers a global region of the object while the heatmap of conv4_x appears in some local salient regions of images.

Based on this, a novel framework, including the multi-scale GeM pooling and the N-pair Center loss, is proposed to combine the coarse features and fine features of an input image. On the one hand, the multi-scale GeM pooling is used to extract the global features and local features respectively. This module consists of two independent channels, each of which includes a residual block of ResNet-18, an attention block and a GeM pooling block. On the other hand, the N-pair Center loss does not need to concern how to select hard or semi-hard negative samples, which makes it faster than the triplet loss and N-pair loss. Furthermore, the framework is end-to-end and all the parameters including GeM pooling are learnable so that the global information and the local information can be ensured to be fused together.

Our main contributions include: Firstly, we discover that the mid-level feature maps as local salient regions could well complement the existing global feature representation. So a new syncretized strategy for the mid-level feature and the global feature is presented. Secondly, we present a novel end-to-end framework with multi-scale GeM pooling and N-pair Center loss. Thirdly, our framework outperforms many state-of-the-art methods in FGIS tasks. our method demonstrates a significant improvement on baselines by up to 2%, especially on the challenging CUB-200-2011 dataset.

The rest of this paper is organized as follows: The section 2 presents the framework with multi-scale GeM pooling and N-pair Center loss. Then we compare our results with the state-of-the-art demonstrating a significant improvement on three datasets in section 3. At last, we reach some conclusions in section 4.

2. PROPOSED METHOD

In this section, we will introduce our method. The framework, as depicted in Fig. 1, consists of two parts: the Multi-scale GeM pooling that is used to aggregate features with attention mechanism and the N-pair Center loss.

2.1. Multi-scale GeM pooling

Previous approaches aggregated the dense extracted features via the max pooling or the average pooling, both of which are unlearnable, while generalized mean (GeM) pooling has a learnable pooling parameter and can be a part of the back-propagation [19]. Concretely, given an input image, feature maps of a residual block is a 3D tensor $\mathcal{X}$ of $K \times W \times H$ dimensions, where $\mathcal{X}_k$ denotes the k-th feature map. Denote $f_n \in R^K, n = 1, \cdots, W \times H$ as K-dimensional local features to be learned jointly with the attention module. The vector $v^{(m)}$ aggregated by max pooling is given by:

$$v^{(m)} = [v_1, v_2, ..., v_K], v_k^{(m)} = \max_{x \in \mathcal{X}_k} x. \quad (1)$$

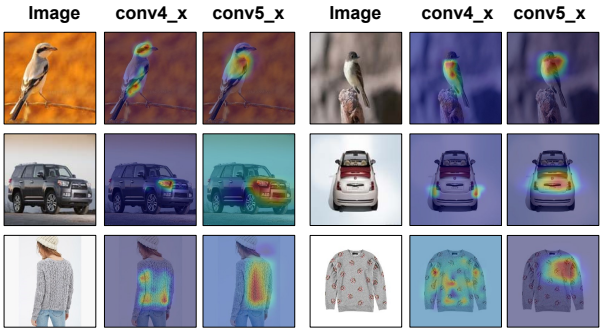

Fig. 2. The output heatmaps of conv4_x and conv5_x generated by class activation mapping on three datasets.

The vector $v^{(a)}$ aggregated by average pooling is given by:

$$v^{(a)} = [v_1, v_2, ..., v_K], v_k^{(a)} = \frac{1}{|\mathcal{X}_k|} \sum_{x \in \mathcal{X}_k} x. \quad (2)$$

The vector $v^{(g)}$ aggregated by GeM pooling is given by:

$$v = [v_1, v_2, ..., v_K], v_k = \frac{1}{|\mathcal{X}_k|} \left(\sum_{x \in \mathcal{X}_k} x^{\mathcal{G}} \right)^{\frac{1}{\mathcal{G}}}. \quad (3)$$

And the max pooling and the average pooling are special cases of GeM pooling, *i.e.* max pooling when $\mathcal{G} \to \infty$ and average pooling for $\mathcal{G} = 1$.

As the retrieval results will be affected by the background of images, especially some image regions are irrelevant to our task,it is crucial to select certain image regions that contain semantic information contributed to the target label. Inspired by DELF [20] and SE-Net [21], we design an effective attention module to choose important regions, the features of which help to distinguish similar categories. Unlike SCDA [7], the proposed attention module selects the pivotal areas from convolutional feature maps during the training period instead of the test period. It directly chooses the key areas by calculating scores for each grid of the convolutional feature maps and can be trained end to end.

The attention module learns a function $g(f; \theta)$ to calculate attention scores for each local features, where θ denotes the parameters of function $g(\cdot) \in [0, 1]$ and the outputs of $g(\cdot)$ have $W \times H$ activations. Given the feature maps $\mathcal{X}$, the generated attention-aware feature maps $\hat{\mathcal{X}}$ are constructed weighted K-dimensional features $\hat{f}_n$ which are given by

$$\hat{f}_n = g(f_n; \theta) \cdot f_n, \quad (4)$$

where $g(\cdot)$ is designed using a 2-layer CNN with a sigmoid activation function at the top and a ReLU activation function at the middle. Note that we employ the convolutional filters of size 1×1.

Fig. 2 shows the heatmaps of conv4_x and conv5_x on the input images that are drawn by the simple technique

CAM [22]. It is obvious that the outputs of conv5_x focus on the overall information of the target in the image while the outputs of conv4_x pay more attention to the part of the target. And both of them are helpful to our task. Based on this, we exploit the feature maps of conv5_x and conv4_x as coarse features and fine features respectively. Concretely, given an input image, the feature maps of conv4_x $\mathcal{X}^{fine}$ and conv5_x $\mathcal{X}^{coarse}$ are taken from ResNet-18. Firstly, these feature maps are multiplied by the scores calculated by attention modules and then aggregated by GeM pooling. Considering the case of $\mathcal{X}^{fine}$, the fine features v^{fine} are aggregated by GeM pooling and the last embedded features y^{fine} are given by

$$y^{fine} = W^{fine} v^{fine}, \tag{5}$$

where $W^{fine} \in R^{E \times K}$ represents the weights of the final embedding layer, E denotes the dimension of the embedded features and K denotes the channels of $\mathcal{X}^{fine}$. As similarly for the coarse features v^{coarse} and the embedded features y^{coarse}. Finally, we use the concatenate of y^{coarse} and y^{fine} as image representation.

2.2. N-pair Center Loss

Triplet loss [15], a simple solution and easy to implement, is one of the most common loss functions in retrieval tasks. A series of methods [8, 16, 23, 18] have been proposed to improve the triplet loss, most of which are based on searching hard negative samples. However, it is a time consuming task to mine hard negative samples during training. In order to accelerate the training process, we propose the N-pair Center loss where N class center features are randomly initialized with L_2-normalization and the output features of the embedding layer are L_2-normalized as well. And the normalization make training more effective [24, 25, 26] Here, the N is equal to the number of training set categories. The N-pair Center loss does not need to concern how to select hard or semi-hard negative samples. Fig. 3 illustrates the concept of the N-pair Center loss that learns a center for each class directly, and then the probability of the data point belonging to the i-th class can be calculated as follow:

$$p_i = \frac{exp\left(-d_i\right)}{\sum_j exp\left(-d_j\right)}, \tag{6}$$

where d_i is the Euclidean distance between the anchor point and the i-th class center. As the features of them are L_2-normalized, the equation (6) can be efficiently calculated by

$$p_i = \frac{exp\left(w_i y\right)}{\sum_j exp\left(w_j y\right)}. \tag{7}$$

Here, it is intuitive to find that the limit of the probability is $\frac{e}{e+\frac{N-1}{e}}$ if the feature y is classified correctly. For instance, the limit is approximately equal to 0.007 when $N = 1000$. It is difficult to be optimized when the categories of training

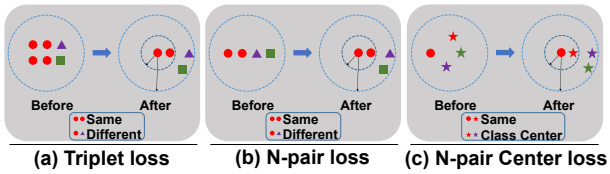

Fig. 3. Illustration of Triplet loss, N-pair loss, and N-pair Center loss in a training batch with four examples. Assuming each color belongs to a different class, the N-pair Center loss in (c) leverages each class center as negative samples. It is faster than the other two losses, because it not need to select negative samples and only compare with the learning class center

set is too large. Considering the fact that the probability of correct classification should be close to 1 and the probability of error should approach 0, we multiply the similarity by a scale parameter s and append a margin parameter m to it. The final probability is given by

$$p_i = \frac{exp\left(sw_i y - m\right)}{exp\left(sw_i y - m\right) + \sum_{j \neq i} exp\left(sw_j y\right)}, \tag{8}$$

Considering the case of $m = 0$, the average probability P of classifying a feature correctly is given by

$$P = \frac{e^s}{e^s + e^{-s} + (N-2)}, \tag{9}$$

Ignoring the term e^{-s}, the probability becomes:

$$P = \frac{e^s}{e^s + N - 2}, \tag{10}$$

if the probability of correct classification is greater than $p \in [0, 1]$, the lower limit of scale parameter is given by

$$s_{low} = log \frac{P(N-2)}{1-P}, \tag{11}$$

where s is approximately equal to 7.5 when $N = 100$ and $P = 0.95$.

For training, the N-pair Center loss is defined as

$$\mathcal{L} = -y^* \cdot log\left(p\right), \tag{12}$$

where y^* is the ground-truth in one-hot representation.

As the proposed framework need to handle the coarse features and fine features, it is inevitable to balance the relationship between these two loss. We use the natural exponential function as the coefficients to balance the relationship and add a L_2 regularization term on λ . The final loss is defined as

$$Loss = e^{-\lambda_1} \mathcal{L}_{fine} + e^{-\lambda_2} \mathcal{L}_{coarse} + \lambda_1^2 + \lambda_2^2. \tag{13}$$

Table 1. Recall@K(%) on the CUB-200-2011 dataset with different embedding sizes and pooling methods

	Avg pooling				Max pooling				GeM			
dimension	R@1	R@2	R@4	R@8	R@1	R@2	R@4	R@8	R@1	R@2	R@4	R@8
512	**54.4**	**66.6**	**76.9**	**85.3**	**55.7**	**67.3**	**77.4**	85.2	54.4	66.6	76.9	85.3
256	53.9	66.0	76.6	84.9	54.1	66.0	77.0	84.9	53.9	66.0	76.6	84.9
128	54.2	65.4	76.3	84.4	55.0	66.9	**77.4**	**85.3**	55.4	67.2	77.4	85.4
64	53.9	66.0	76.2	84.8	53.9	66.0	76.3	84.1	**56.2**	**68.3**	**78.1**	**86.1**

Table 2. Recall@K(%) on the CUB-200-2011 dataset with feature maps extracted by different layers

layer name	R@1	R@2	R@4	R@8
conv5_x	**55.0**	**66.9**	**77.4**	**85.3**
conv4_x	38.1	50.3	61.9	72.7
conv3_x	15.8	24.5	33.9	45.3

3. EXPERIMENTAL RESULTS

3.1. Experimental setup

Datasets: We use the same train/test setting as [11] for all datasets. There are 200 classes of birds with 11788 images in the CUB-200-/2011 dataset [12]. We use the first 100 classes for training (5864 images) and the rest for testing (5924 images). The CARS196 dataset includes 16185 car images with 196 classes totally [27]. According to the settings, first 98 classes are used for training(8054 images) and the rest are used for testing(8131 images). There are 7982 classes of clothing items with 52687 images in the In-shop Clothes Retrieval dataset [8]. We use the first 3997 classes for training (25866 images) and the rest for testing (26821 images). The test set is partitioned into the query set containing 3985 classes with 14213 images and the gallery set containing 3985 classes with 12608 images. Given a query image in the test set, we retrieve the most similar image in the gallery set.

Evaluation Protocols: Following the experimental protocol in DREML [11], we evaluate the retrieval results by the standard metric Recall@K. Recall@K is the average recall scores over all query images in the test datasets including untrained classes from the CUB-200-2011, CARS196, and In-shop Clothes Retrieval datasets. Concretely, the top K similar images are returned for each query image. The recall score will be 1 if there is at least one positive image in the top K returned images, and 0 otherwise.

Baselines: We compare our method with four current state-of-art methods: Boosting Independent Embeddings Robustly (BIER) [10] divides the last embedding layer of a deep network into an embedding ensemble. An Adversarial Approach to Hard Triplet Generation (HTG) [23] aims at generating those hard triplets rather than mining hard triplets by an adversarial approach. Deep Randomized Ensembles for Metric Learning (DREML) [11] uses multiple networks with same architecture that are trained on datasets with different kinds of partitions. Hard-Aware Deeply Cascaded Embedding (HDC) [18].

Implementation Details. We choose the ResNet-18 [14] pre-trained on ImageNet ILSVRC-2012 [13] as the skeleton network. For all training datasets, the input images are first resized to 281 by 281 pixels, then cropped at 256 by 256 pixels. For the data augmentation scheme, we use random rotation within 10 degrees, random horizontal flip and random crop for training, and apply a single center crop for testing. For preprocessing, we normalize the images using the channel means and standard deviations following the setup in [11]. All experiments are trained by using stochastic gradient (SGD) with the momentum set to be 0.9. We employ the hyperparameters without specific tuning, the initial learning rate set to 1e-3, the weight decay set empirically to 5e-4, the scale and margin parameters set to be 16 and 4 respectively. The balance coefficients λ_1 and λ_2 of final loss function are initialized with 0. The mini-batch size is 64 for CUB-200-2011 and CARS196 datasets and 128 for In-shop Clothes Retrieval dataset.

3.2. Parameter setting

We demonstrate the improvements of our multi-layer framework, then compare the results that only contain the feature maps of conv5_x taken from ResNet-18 without any additional branch. As shown in Table 2, the features extracted from the outputs of conv4_x and conv5_x have better discriminative ability, which are consistent with the performance in Fig. 2 as well. The lower-level convolution kernel extracts specific texture information, the higher-level convolution kernel extracts the combination of colour and texture features which contain more abstract target features. This is the reason why the framework only use the last two blocks of the ResNet-18.

The generalized mean pooling, max pooling and average pooling are the most general pooling approaches. To evaluate these three pooling methods, we do a group of experiments on CUB-200-2011 dataset via the features extracted from the feature maps of conv5_x taken from ResNet-18. Table 1 shows the performance tradeoffs among different choices of the dimension of embeddings and pooling methods, which indicates that the generalize mean pooling is better than other two methods in different embedding sizes. In addition, it can be seen from Tabel 1 that the embedding size has little effect on the experiment results. In order to save storage space, we set the embedding size to be 128 in the following experiments.

Table 3. The retrieval performance on the CUB-200-2011, CARS196 and In-shop Clothes Retrieval datasets

	CUB-200-2011				CARS196				In-shop Clothes Retrieval			
Methods	R@1	R@2	R@4	R@8	R@1	R@2	R@4	R@8	R@1	R@10	R@20	R@30
HDC [18]	53.6	65.7	77.0	85.6	73.7	83.2	89.5	93.8	62.1	84.9	89.0	91.2
BIER [10]	55.3	67.2	76.9	85.1	78.0	85.8	91.1	95.1	76.9	92.8	95.2	96.2
HTG [23]	59.5	71.8	81.3	88.2	76.5	84.7	90.4	94.0	80.3	**93.9**	**95.8**	96.6
DREML [11]	63.9	75.0	83.1	89.7	**86.0**	**91.7**	**95.0**	**97.2**	78.4	93.7	**95.8**	**96.7**
Ours	**66.0**	**75.8**	**84.4**	**90.6**	85.0	91.0	94.4	96.6	**82.1**	**93.9**	95.6	96.3

Fig. 4. The retrieval results on the CUB-200-2011, CARS196, In-shop Clothes Retrieval dataset. We retrieve the most 4 similar images to the query image. The correct results are highlighted in green and the incorrect results are highlighted in red.

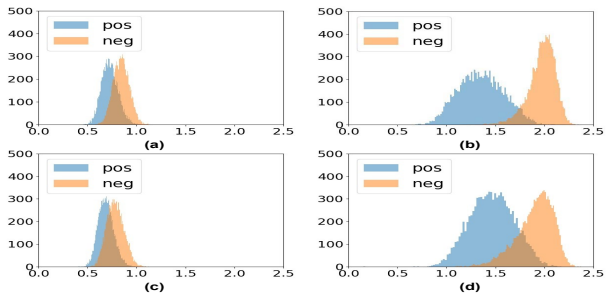

Fig. 5. Histograms for positive and negative distance distributions of randomly selected 10000 images on the CUB-200-2011 train/test set for: (a)/(c) Initial state: randomly initialized net, (b)/(d) Network training with N-pair Center loss.

3.3. Quantitative results

Table 3 shows the retrieval performance on CUB-200-2011, CARS196 and In-shop Clothes Retrieval datasets. The experiments on the CUB-200-2011 and In-shop Clothes Retrieval datasets demonstrate a significant improvement on state-of-the-art results by up to 2% on both datasets. For CUB-200-2011 and CARS196 datasets, both of them contain substantial intra-class variability. Fig. 4 shows example retrieval results of our method on three datasets and (left) highlights the pose variations of objects and the diversity of backgrounds. In CUB-200-2011 dataset, birds are shown in different poses with different outdoor backgrounds. As for In-shop Clothes Retrieval dataset, it has more categories than CUB-200-2011 and CARS196 datasets, and almost includes no more than ten samples for per class. For the CARS196 dataset, it tend to differ in smaller parts than CUB-200-2011 dataset. And our method has a close performance as the best ensemble model that uses multiple networks on CARS196 dataset, but our framework has less parameters and shorter embedding size.

The embedding distribution of the positive and negative pairs of randomly selected 10000 images from unseen CUB-200-2011 test set is depicted in Fig. 5. The result shows that the N-pair Center loss makes the distance distribution of positive and negative pairs less overlapping than the pre-trained ResNet-18 network, and also demonstrates that our method can effectively spread out the representation.

4. CONCLUSIONS

We present a novel framework including multi-scale GeM pooling and N-pair Center loss. The framework extracts the compact coarse features and fine features with 128-dim respectively, and then combines them together as the image representation. The experimental results reveal that our method outperforms favourably compared to four baseline models on three datasets. We show that the features extracted by our method have better robustness than baseline methods.

5. ACKNOWLEDGMENTS

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